State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains' annual survey of 23,262 developers worldwide

Covering languages, tools, AI attitudes, and career trends

The most comprehensive snapshot of the global developer community

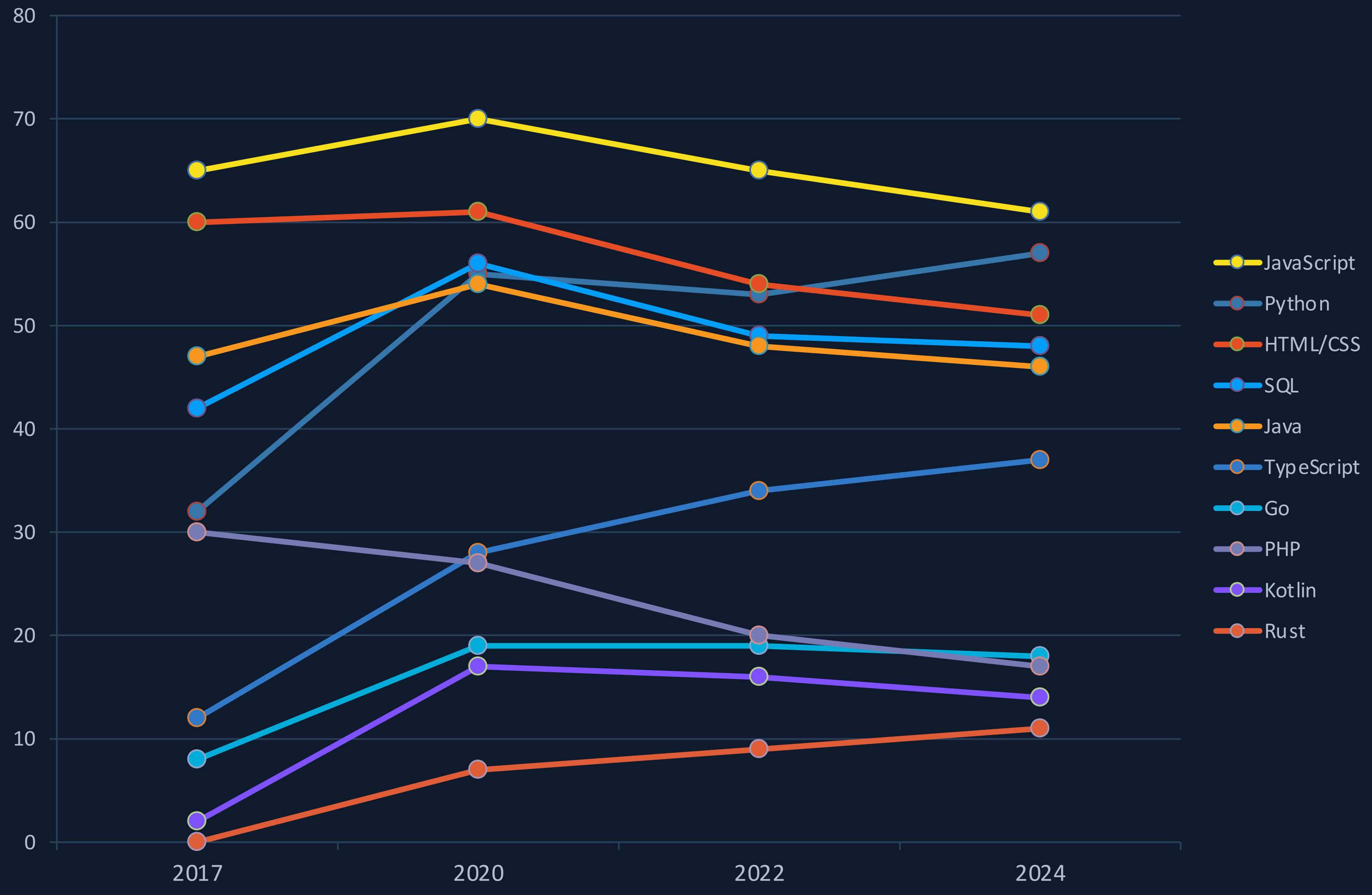
23,262
Developers Surveyed

8 Years
Trend Data (2017-2024)

20+
Languages Tracked

Python Closes the Gap: 8 Years of Language Usage Shifts

Percentage of developers using each language | 2017-2024



JavaScript

Peaked at 70% in 2020, now declining to 61%

Python

Rose from 32% to 57%, nearly closing the gap

TypeScript

Tripled from 12% to 37%, strongest growth trajectory

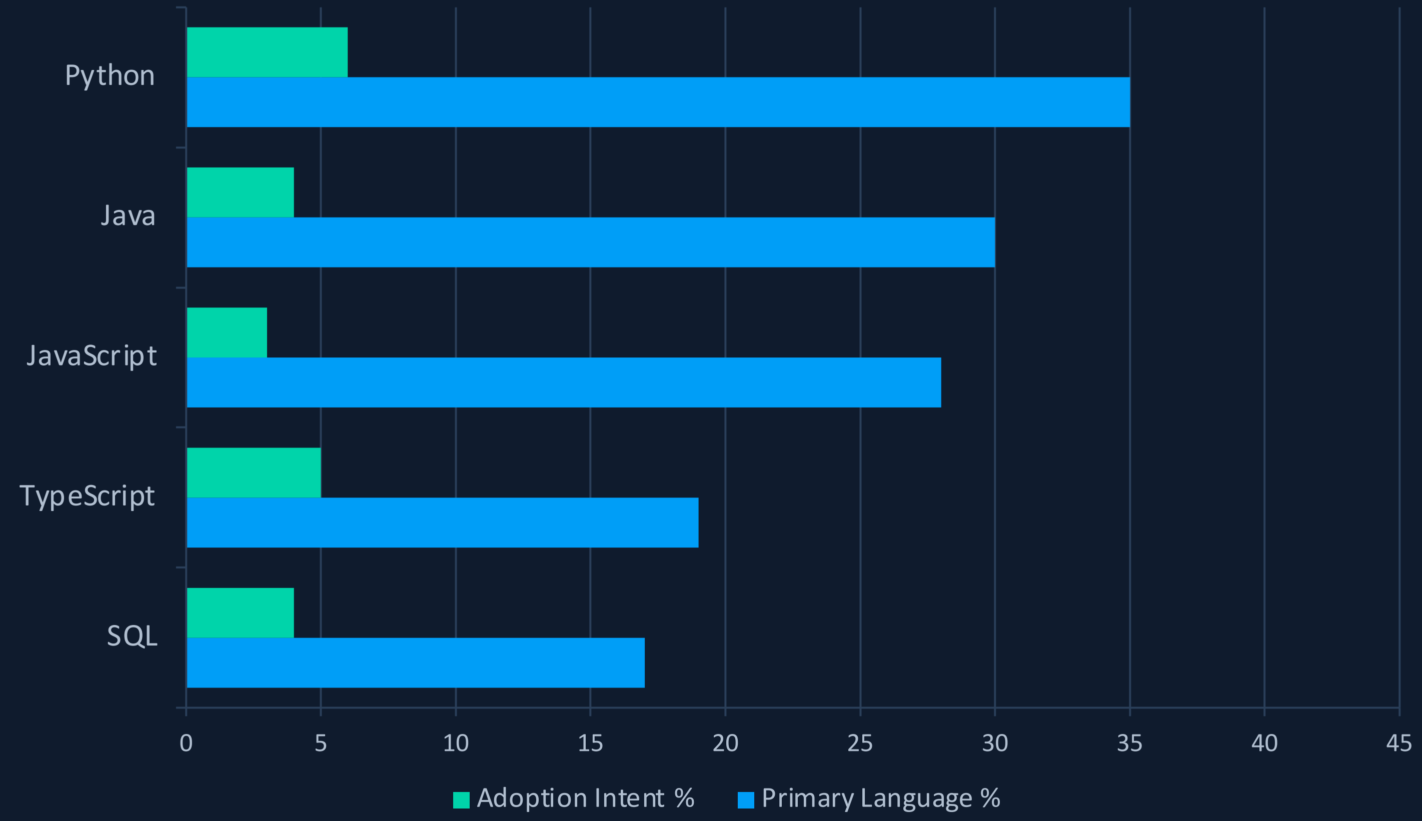
PHP

Dropped from 30% to 17%, significant long-term decline

Source: JetBrains State of Developer Ecosystem 2024 | n = 23,262

Python Is Now the #1 Primary Language at 35%

Primary language selection vs. adoption intent — developers stick with what they know



KEY INSIGHT

The Comfort Zone Effect

Adoption intent stays below 6% across all languages — developers rarely plan to switch their primary stack.

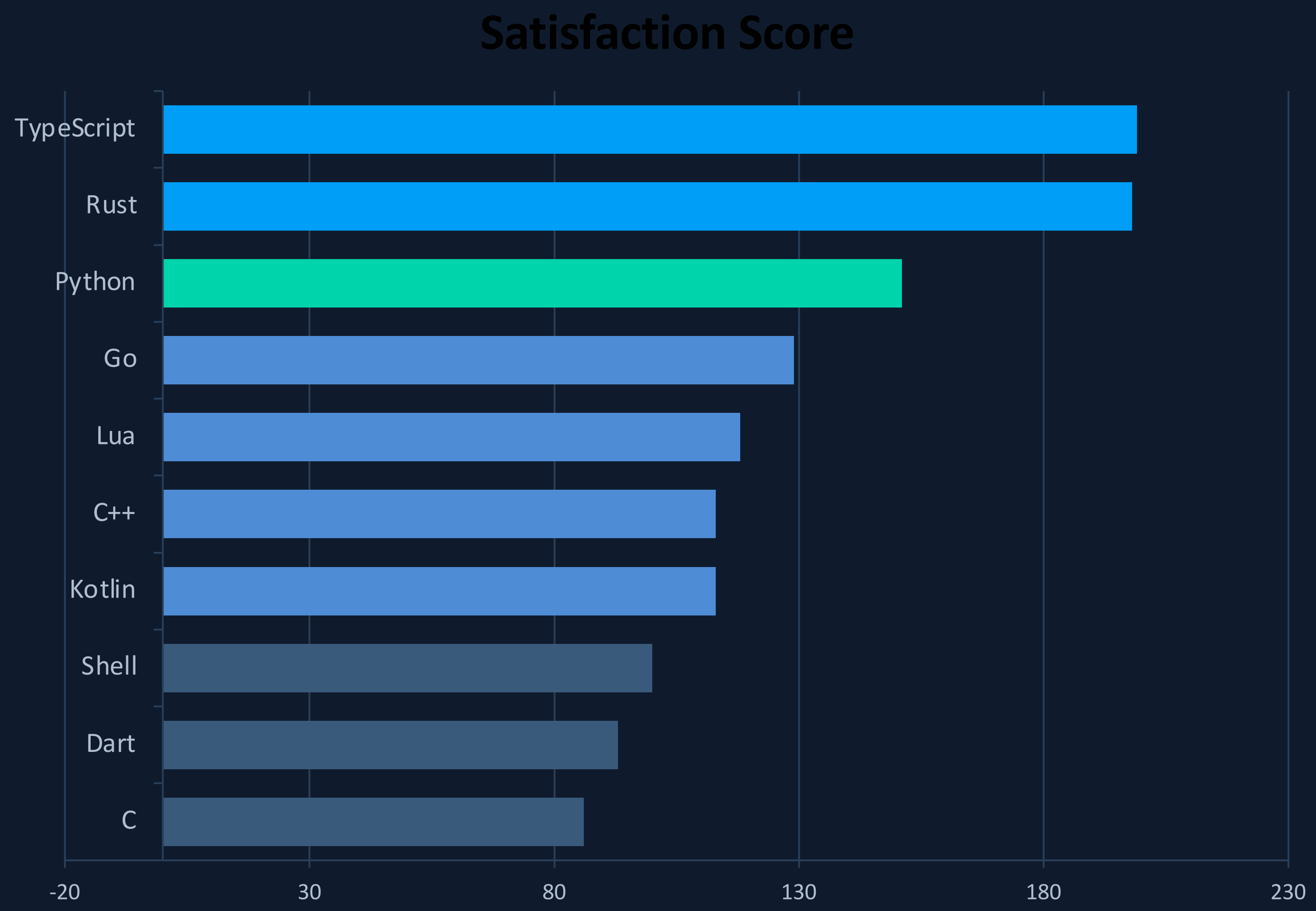
Python's 35% primary language share signals that AI/ML growth has converted casual users into committed practitioners.

JavaScript remains the most used language overall (61%) but only 28% call it their primary — indicating breadth over depth.

Ecosystem lock-in and expertise investment keep developers loyal to their chosen stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Language satisfaction scores — type-safe, modern-tooling languages dominate the top spots



WHAT DRIVES SATISFACTION?

Type-safe systems lead

TypeScript and Rust combine compiler safety, expressive type systems, and modern tooling — the common thread among top-scorers.

Python's breadth advantage

Python scores 151 — strong satisfaction buoyed by AI/ML and data science momentum.

SATISFACTION TIERS

- 190+ Elite — TypeScript, Rust
- 140+ High — Python
- 110+ Good — Go, Lua, C++, Kotlin
- <110 Average — Shell, Dart, C

Source: JetBrains State of Developer Ecosystem 2024 | n = 23,262

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by domain, not general preference — each ecosystem has its stronghold

Application Domain	Leading Language	Share	Visual
AI / Machine Learning	**Python**	**31%**	
Data Science & Analytics	**Python**	**28%**	
Data Processing & Analytics	**Python**	**17%**	
Integrating with APIs & Services	**Python**	**38%**	
User Interface	**TypeScript**	**62%**	
APIs & Services	**Java**	**49%**	
Application Logic & Workflows	**Java**	**55%**	
System Tools & Components	**C++**	**27%**	

Note: User Interface development also sees JavaScript at 58% — nearly matching TypeScript's 62%, reflecting the ongoing JS-to-TS migration.

Hardware Interfacing (C++ 21%) and Graphics Rendering (C++ 15%) further reinforce C++'s systems domain.

Source: JetBrains State of Developer Ecosystem 2024 | n = 23,262

What This Means for Your Technology Strategy in 2025

Five actionable insights for engineering leaders and hiring managers

1 Invest in Python for AI/ML Teams

Python is the undisputed leader in AI/ML (31%) and data science (28%). Its rise from 32% to 57% usage correlates directly with the AI boom. Any AI/ML hiring strategy must center on Python expertise.

35% primary language — #1 rank

2 TypeScript Is the Future of Frontend

TypeScript leads UI development at 62% and tops developer satisfaction at 199. Tripling from 12% to 37% usage, it is displacing JavaScript as the default for frontend and full-stack work.

3x growth in 8 years | #1 satisfaction

3 Rust and Go Are the Infrastructure Growth Plays

Both address modern infrastructure needs — microservices, cloud-native tooling, systems programming — where performance and reliability are non-negotiable. Rust (satisfaction 198) and Go (stabilized at 18%) are strategic bets for platform teams.

Rust: 0% to 11% | Go: 8% to 18%

4 PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Both are losing ground to newer alternatives. Organizations relying on these stacks should plan migration paths and diversify hiring.

PHP -43% | Ruby -60% since 2017

5 Satisfaction Correlates with Type Safety — Factor into Retention

The top satisfaction spots belong to TypeScript (199) and Rust (198), both combining type safety with modern tooling. Developer experience directly impacts retention — investing in type-safe, well-tooled stacks is a retention strategy.

Top 2 satisfaction: both type-safe languages

Source: JetBrains State of Developer Ecosystem 2024 | n = 23,262